AgriCapsNet: An Attention-Aware Capsule Network for Multicrop Agricultural Landscape Classification Using Hyperspectral Imagery

Wei Zhang, Yupu He, Zhaohui Xue, *Member, IEEE*, Shanchuan Guo, Peng Zhang, Cong Lin, and Wei Qi

***Abstract*—Fine classification of complex agricultural landscapes is essential for precise agricultural information acquisition. Hyperspectral imagery (HSI) combined with convolutional neural networks (CNNs) offers an effective means for land cover classification. However, conventional CNNs are built on scalar neurons, which limits their ability to effectively filter and capture joint spectral–spatial information in complex scenes. To overcome these limitations, this study proposes the agricultural capsule network (AgriCapsNet), an encoder–decoder capsule network designed for Multicrop agricultural landscape classification with HSI. The encoder integrates a feature-focused former module, comprising lightweight convolution and attention mechanisms, to enhance feature refinement and extraction, together with capsule layers that leverage vector neurons to strengthen hierarchical feature representation. The decoder employs a joint margin and reconstruction loss to improve classification generalization. In addition, a new hyperspectral dataset, Zhang-Jia-Gang (ZJG), is publicly released to support research on fine-grained agricultural landscape classification. AgriCapsNet was evaluated on three crop-rich HSI datasets and outperformed nine competing models, achieving overall accuracy (OA) improvements of 0.90%–24.49%, 1.87%–28.82%, and 1.84%–7.09%, respectively. It also demonstrated strong robustness to noise and adaptability to different pixel neighborhood sizes, confirming its suitability for large-scale agricultural monitoring.**

***Index Terms*—Agricultural landscape classification, attention mechanism, capsule network, hyperspectral imagery (HSI), lightweight convolution.**

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I. Introduction

TIMELY and reliable agricultural information is crucial for effective grain market operations, policymaking, and food security [1]. Rapid and accurate classification of diverse crop cultivation areas is therefore essential for maintaining social stability and promoting sustainable development. This task becomes particularly challenging in complex agricultural landscapes, which are often fragmented and contain a wide variety of crop types and intercropping patterns. Moreover, environmental factors such as soil background variability and weed infestation further complicate spectral characteristics, increasing intra-class variability and reducing inter-class separability.

Hyperspectral imagery (HSI) offers unique advantages for agricultural landscape classification due to its ability to precisely characterize spectral properties [2], [3]. In recent years, extensive research has explored HSI-based classification methods for agricultural applications, including crop type mapping, intercropping identification, and fine-scale field-level monitoring [4], [5], [6], [7], [8]. These studies demonstrate that rich spectral information can effectively enhance crop discriminability, particularly in heterogeneous agricultural environments. However, practical applications of HSI still face challenges such as data redundancy, mixed-pixel effects, and the Hughes phenomenon, which may reduce classification accuracy and hinder large-scale adoption [4]. To improve HSI-based classification, previous research has mainly focused on three aspects: enhancing data quality, optimizing feature representation, and designing advanced classifiers. Data quality enhancement typically involves image preprocessing and sample library construction [9], [10]; feature optimization includes spatial- and frequency-domain operators, sparse representation, and deep learning-based feature extraction [3], [11]; and classifier design encompasses improved traditional machine learning algorithms, multiclassifier ensembles, and deep neural networks [12], [13], [14], [15].

Combining optimized features with machine learning algorithms has become a widely adopted paradigm for HSI classification. Existing approaches can generally be grouped into three categories. First, spectral-based classification analyzes spectral differences among targets using spectral matching techniques such as spectral information divergence (SID) and spectral angle mapper (SAM) [16]. However, such

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methods typically neglect spatial context, often resulting in speckled classification artifacts. Second, discriminative spatial-spectral features have been designed, including extended morphological profiles (EMPs) [17], extended attribute profiles (EAPs) [18], Gabor operators [19], and HGFM [20], which are then combined with classifiers such as random forest (RF) and support vector machine (SVM) [21], [22]. Third, improvements to traditional machine learning frameworks have been explored, for example, rotation forest [23] and optimized spatial–spectral conditional random fields (CRFs) [24]. Although these strategies can enhance classification accuracy, hand-crafted features often remain limited in adaptability and effectiveness when applied to heterogeneous cultivated landscapes.

With the rapid advancement of deep learning techniques [25], [26], HSI classification has achieved substantial progress in the agricultural remote sensing community [27], [28]. Commonly adopted models include convolutional neural networks (CNNs), recurrent neural networks (RNNs), stacked autoencoders (SAEs), and deep belief networks (DBNs). Among them, CNN-based architectures are particularly effective at extracting spatial features, with variants such as 1D-CNN, 2D-CNN, and 3D-CNN, demonstrating promising classification performance [27], [29]. Nevertheless, several issues persist. Models such as RNNs and DBNs often require converting spatially structured imagery into 1-D vectors, which may lead to the loss of joint spatial-spectral information. In addition, conventional CNNs built on scalar neurons have limited capacity to represent complex feature relationships and often lack robustness in capturing rotational invariance.

More recently, advanced deep learning architectures and hyperspectral foundation models have further driven progress in HSI classification. Transformer-based models such as DSFormer exploit self-attention mechanisms to model long-range spectral–spatial dependencies and improve feature representation [30]. Some studies also address practical challenges in hyperspectral annotation and multimodal learning. For instance, the MAGS network integrates partial label learning with a Siamese reconstruction structure to alleviate label ambiguity [31], while the MMD-MLP model incorporates multiresolution, multiscale feature extraction together with LiDAR-guided enhancement to strengthen hyperspectral feature representation [32]. Furthermore, newly proposed hyperspectral large models, including HyperSIGMA [33] and HyperFree [34], aim to learn more generalizable spectral representations through large-scale pretraining, adaptive channel modeling, and tuning-free transfer capabilities. These approaches demonstrate strong potential for improving robustness and generalization in hyperspectral image understanding, particularly when labeled samples are limited or scene conditions are complex. Despite these advances, effectively modeling hierarchical spatial relationships while preserving structured spectral–spatial information in fragmented agricultural landscapes remains challenging.

The concept of capsules was first introduced by Hinton et al. [35], and the capsule network (CapsNet) was formally proposed in [36]. By employing dynamic routing and vectorized neurons, CapsNet can capture hierarchical spatial relationships and model the instantiation parameters of object features [36], [37]. To better capture relative positional relationships among emerging features, Paoletti et al. [38] pioneered the application of CapsNet to HSI classification, achieving improved performance. Subsequently, CapsNet has been explored through adaptive adjustment [39], [40], architectural modifications [41], [42], and local optimization [43], [44], demonstrating competitive results. However, leveraging CapsNet to capture joint spectral–spatial information in complex agricultural landscapes remains challenging. Conventional capsule networks are not specifically designed for agricultural scenarios and lack mechanisms to suppress background noise and emphasize meaningful spectral–spatial patterns, leading to suboptimal feature extraction under mixed pixels and multiple crop types. Moreover, standard capsule layers may struggle to model hierarchical spatial relationships in fragmented fields, limiting discriminative capability and generalization across varying environmental conditions.

To address these challenges, this study proposes the agricultural capsule network (AgriCapsNet), an encoder–decoder capsule network for fine classification of cultivated areas using HSI. The encoder integrates attention mechanisms and lightweight convolutions to strengthen spatial–spectral feature extraction, while the capsule layers enhance hierarchical feature representation. The decoder employs a joint margin-reconstruction loss to improve classification generalization. Compared with baseline methods, AgriCapsNet achieves higher accuracy and efficiency in Multicrop agricultural landscape classification, demonstrating strong robustness and scalability across complex environmental conditions.

The remainder of this article is organized as follows. Section II describes the proposed model in detail. Section III presents the datasets, experimental setup, and classification results. Section IV discusses the findings, and Section V concludes this article.

II. Methodology

A. Overview of the Proposed Model

Agricultural hyperspectral image classification is characterized by complex spectral–spatial patterns, including high spectral similarity among crops, pervasive mixed pixels, and fragmented field structures. Although capsule-based networks have demonstrated advantages in modeling part–whole relationships and preserving spatial hierarchies, most existing capsule models are primarily designed for object-centric scenes and natural images. As a result, they often struggle to effectively capture fine-grained spectral–spatial variations and contextual dependencies that are critical in agricultural hyperspectral scenarios.

To address these challenges, AgriCapsNet is specifically designed from an agricultural perspective and incorporates three targeted improvements. First, a spatial-channel attention mechanism is introduced to adaptively emphasize informative spatial regions and discriminative feature channels, thereby alleviating background interference and enhancing subtle crop-related features. Second, capsule representations are employed to better model local spectral–spatial feature compositions

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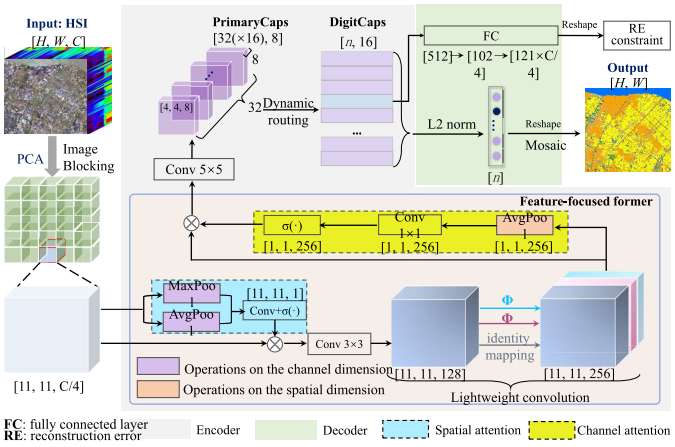


Fig. 1. Overall architecture of AgriCapsNet.

under mixed-pixel and fragmented conditions. Third, the overall architecture is optimized to improve feature aggregation and robustness across different agricultural scenes, enabling more effective discrimination among spectrally similar crops.

The overall architecture of AgriCapsNet is illustrated in Fig. 1, which primarily consists of an encoder and a decoder. The encoder is designed to enhance the model's ability to capture and recognize spatial–spectral features, thereby improving the network's focus and representational capacity. It comprises a feature-focused former and capsule layers, where the capsule component includes the PrimaryCaps and DigitCaps. The decoder is mainly responsible for identifying land cover types within cultivated areas, and it performs two tasks: reconstruction and classification. By jointly optimizing reconstruction loss and classification loss, the model's generalization capability can be effectively improved.

The specific modules and computational processes involved in both the encoder and decoder are detailed in Sections II-B–II-D.

B. Feature-Focused Former

As part of the encoder, the feature-focused former is primarily composed of convolutional operators, lightweight convolution modules, and attention mechanisms.

1) Convolutional Operator: The core of a convolutional operator lies in the learnable convolution kernels, which perform convolution operations on the input from the previous layer (either an image or feature map) and progressively extract deeper features. To effectively capture multi-level, multi-scale, and multi-type features from the input data, a single convolutional layer typically contains multiple convolution kernels, producing multiple output feature maps. For a standard convolution, the computation of a single output feature map can be expressed as [45]

$$\begin{cases} u_\beta^l = f\left(v l_\beta\right) \\ v_\beta^l = \sum_{\alpha \in A_\beta} u_\alpha^{l-1} * \kappa_{\alpha\beta}^l + b_\beta^l. \end{cases} \tag{1}$$

Here, u_β^l denotes the activation output of the βth channel in the lth convolutional layer, the function $f(\cdot)$ represents the activation function, which is typically selected from Tanh, Sigmoid, or ReLU [46]. The v_β^l refers to the pre-activation feature in the βth channel of layer l, obtained by the summation of convolution operations between the feature maps u_α^{l-1} from the previous layer and the corresponding convolution kernel $\kappa_{\alpha\beta}^l$, followed by the addition of a bias term b_β^l. A_β denotes the subset of input feature maps used to compute v_β^l, and "$*$" represents the convolution operator.

2) Lightweight Convolution Module: CNNs are commonly used to capture spatial features, and typically consist of multiple convolutional layers with a large number of trainable parameters. To effectively reduce computational cost while extracting feature representations that are comparable to or even more diverse than those obtained by standard convolutional networks, a lightweight convolution module is constructed in this work based on the Ghost module [47]. Specifically, a standard convolution is first applied to the input features, followed by a depthwise convolution, which can be regarded as a linear operation. The main computational process is described in the following equations:

$$Y = u * \kappa \tag{2}$$

where Y denotes the intrinsic feature obtained after the standard convolution, u represents the input feature, and the convolution kernel $\kappa \in \mathrm{R}^{m\times k\times k\times c}$

$$y_{ij} = \Phi_{i,j}(y_i) \quad \forall i = 1, .., m, \ \ j = 1, \ldots, s. \tag{3}$$

Here, $\Phi_{i,j}$ represents the jth linear operation applied to the ith intrinsic feature y_i of Y in (2), which generates the corresponding jth feature map y_{ij}. The last operation $\Phi_{i,s}$ denotes the identity mapping applied to the intrinsic feature. The final set of feature is given by $\{y_{11}, y_{12}, ..., y_{ms}\}$. The relationship among the number of intrinsic features m, the number of linear operations s, and the total number of output feature channels n can be expressed as $n = m \cdot s$ ($m \le n$). Excluding the last identity mapping, this relationship can be further formulated as $m(s-1) = n/s \cdot (s-1)$.

To illustrate how the lightweight convolution module reduces computational cost, we take the theoretical speed-up ratio and parameter compression ratio as examples. Assume the input feature u has dimensions $h \times w \times c$ (height $\times$ width $\times$ channels), and the output feature map has dimensions $h' \times w' \times n$. For a standard convolution, let the convolution kernel κ have a size of $k \times k \times c$ (height $\times$ width $\times$ input channels). The corresponding floating-point operations (FLOPs) are given by

$$FLOPs_{(\mathrm{conv})} = h' \cdot w' \cdot n \cdot k \cdot k \cdot c. \tag{4}$$

Let the size of the linear operation kernel in the lightweight convolution module be $d \times d$. Then, the FLOPs of the lightweight convolution module can be expressed as

$$\begin{aligned} FLOPs &= m \cdot h' \cdot w' \cdot k \cdot k \cdot c + (s-1) \cdot m \cdot h' \cdot w' \cdot d \cdot d \\ &= \frac{n}{s} \cdot h' \cdot w' \cdot k \cdot k \cdot c + (s-1) \cdot \frac{n}{s} \cdot h' \cdot w' \cdot d \cdot d. \end{aligned} \tag{5}$$

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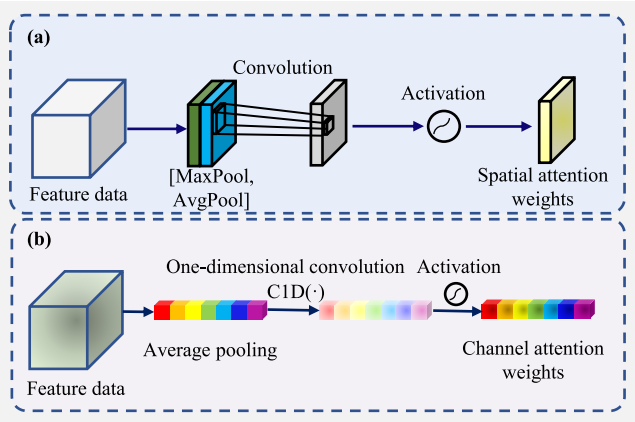


Fig. 2. Workflow of (a) spatial and (b) channel attention modules.

Based on the FLOPs derived above, the theoretical speed-up ratio r_s of the lightweight convolution module relative to standard convolution can be further expressed as

$$\begin{aligned} r_s &= \frac{FLOPs_{(\text{conv})}}{FLOPs} \\ &= \frac{n \cdot h' \cdot w' \cdot k \cdot k \cdot c}{\frac{n}{s} \cdot h' \cdot w' \cdot k \cdot k \cdot c + (s-1) \cdot \frac{n}{s} \cdot h' \cdot w' \cdot d \cdot d} \\ &= \frac{k \cdot k \cdot c}{\frac{1}{s} \cdot k \cdot k \cdot c + (s-1) \cdot \frac{1}{s} \cdot d \cdot d} \approx \frac{s \cdot c}{c + s - 1} \approx s \end{aligned} \tag{6}$$

where d is assumed to be approximately equal to k, and $s \ll c$.

Similarly, the parameter compression ratio r_c of the lightweight convolution module relative to standard convolution can be expressed as

$$r_c = \frac{n \cdot k \cdot k \cdot c}{\frac{n}{s} \cdot k \cdot k \cdot c + (s-1) \cdot \frac{n}{s} \cdot d \cdot d} \approx s. \tag{7}$$

As shown in (6) and (7), when generating output feature maps of the same size, the computational cost of the lightweight convolution module is only $1/s$ that of standard convolution. This effectively reduces the overall computational burden of the network and lowers the model complexity. In the proposed module, the value of s is set to 2.

3) Attention Mechanisms: To enhance the discriminative capability of feature representations, an attention mechanism is introduced to adaptively emphasize informative spatial regions and feature channels. As illustrated in Fig. 2, the proposed attention mechanism consists of two complementary components: a spatial attention module and a channel attention module. The former focuses on identifying important spatial locations, while the latter models inter-channel dependencies to highlight informative spectral features.

Inspired by the convolutional block attention module (CBAM) [48], a spatial attention module is constructed. Specifically, the input feature data is first compressed along the channel dimension by applying both max pooling and average pooling. The two extracted feature maps are then concatenated, followed by a convolution operation and activation. The detailed computation is formulated as

$$\boldsymbol{O}_s = \sigma\left(\text{Conv}\left(\left[\text{AvgPool}\left(\mathbf{F}\right); \text{MaxPool}\left(\mathbf{F}\right)\right]\right)\right). \tag{8}$$

Here, $\boldsymbol{F}$ denotes the input feature data; AvgPool(·) and MaxPool(·) represent average pooling and max pooling operations, respectively. The Conv(·) denotes the convolution operation with a kernel size of 7×7, $\sigma(\cdot)$ is the Sigmoid activation function, and O_s represents the spatial attention weights.

The Channel attention module is implemented using an adaptive 1-D convolution to achieve cross-channel interaction without dimensionality reduction. The specific design is inspired by ECA-Net [49]. First, spatial compression is performed on the input data by applying average pooling. Then, information exchange between the current channel and its k neighboring channels is considered through a 1-D convolution, as expressed in the following equation:

$$\boldsymbol{\omega} = \sigma\left(\text{C1D}_k\left(\boldsymbol{y}\right)\right) \tag{9}$$

where y denotes the input channel feature, C1D(·) represents the 1-D convolution, $\sigma(\cdot)$ is the Sigmoid activation function, and $\boldsymbol{\omega}$ denotes the obtained channel attention weights. The kernel size k of the 1-D convolution is adaptively determined by

$$k = \psi\left(C\right) = \left|\frac{\log\left(C\right)}{\gamma} + \frac{b}{\gamma}\right|_{\text{odd}} \tag{10}$$

where C denotes the number of input channels and k represents the size of the 1-D convolution kernel. Here, the parameters are set as $b = 1$, $\gamma = 2$.

C. Capsule Layer

The capsule layer consists of PrimaryCaps and DigitCaps. PrimaryCaps is responsible for transforming low-level features extracted by convolutional kernels into vector-form capsules. It comprises multiple parallel small convolutional kernels, each extracting a group of features to form a capsule vector. DigitCaps further processes the output of PrimaryCaps through dynamic routing, generating high-level capsules that represent the semantic attributes of each class. Capsule networks determine whether an object belongs to a target class by integrating the presence probability of feature instances with their pose information. In this architecture, vector-based neurons represent capsules, mimicking the object recognition mechanism of the human visual cortex. The magnitude and orientation of each vector respectively encode the probability of an entity's existence and its pose attributes [36]. The computation in capsule networks mainly involves the information transmission between vector neurons and dynamic routing.

1) Information Transmission Between Vector Neurons: The forward propagation process of information transmission between vector neurons (also called capsules) in capsule networks is illustrated in Fig. 3. Unlike traditional neural networks that use scalar neurons for forward information flow, capsule networks transmit information in the form of capsules [36]. The main process consists of three parts: affine transformation, weighted summation, and nonlinear activation.

The affine transformation encodes the abstract spatial relationship between lower-level capsules and higher-level features. Specifically, the input lower-level capsules are multiplied

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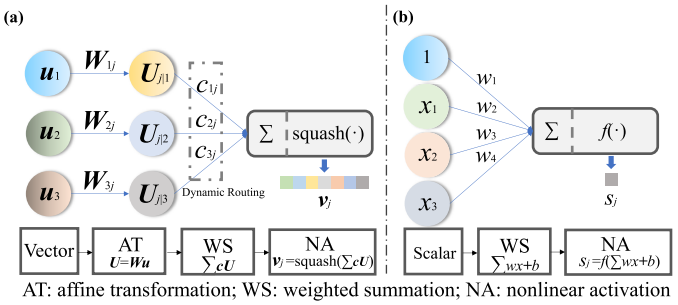


Fig. 3. Example of information forward propagation of neurons in capsule and traditional neural network. (a) Forward propagation of capsule vector neurons. (b) Forward propagation of scalar neurons.

by the weight matrices $\boldsymbol{W}_{ij}(i = 1, 2, 3,...)$ to obtain higher-level features. Weighted summation uses coupling coefficients c_{ij} $(i = 1, 2, 3,...)$ to weight and sum the higher-level features, producing the input vectors required by the higher-level capsules. The specific computation formula is as follows:

$$\begin{cases} \boldsymbol{U}_{j|i} = \boldsymbol{W}_{ij}\boldsymbol{u}_i \\ \hat{\boldsymbol{v}}_j = \sum_i c_{ij}\boldsymbol{U}_{j|i} \end{cases} \tag{11}$$

where $\boldsymbol{u}_i$ denotes the input lower-level capsule, $\boldsymbol{W}_{ij}$ is the weight matrix, $\boldsymbol{U}_{j|I}$ represents the higher-level feature, the coupling coefficient c_{ij} is non-negative and satisfies the constraint that the sum over j equals 1, and $\hat{v}_j$ denotes the input vector required by the higher-level capsule.

The nonlinear activation employs the squash function to normalize the length of the vector $\hat{v}_j$ to the range (0, 1) while preserving its direction. Consequently, the magnitude of the final output higher-level capsule $\boldsymbol{v}_j$ can be used to represent the probability of the target object's presence, as follows:

$$\boldsymbol{v}_j = \frac{\left\|\hat{v}_j\right\|^2}{1+\left\|\hat{v}_j\right\|^2}\frac{\hat{v}_j}{\left\|\hat{v}_j\right\|}. \tag{12}$$

Here, $\|\cdot\|$ denotes the L2 norm, i.e., the magnitude of the vector, and $\|\cdot\|^2$ represents the square of the L2 norm. The first part of the formula performs the squashing of the vector magnitude, while the latter part preserves the direction of the vector. Overall, the operation normalizes the vector while maintaining its original direction.

2) Dynamic Routing: The primary purpose of dynamic routing is to update and determine the coupling coefficients c_{ij}, thereby addressing how to selectively transfer target feature information from lower-level capsules to higher-level capsules with appropriate weights. The operation of the dynamic routing is illustrated in Fig. 4. The input lower-level capsule $\boldsymbol{u}_i$ is multiplied by the weight matrix $\boldsymbol{W}_i$ to obtain the higher-level feature $\boldsymbol{U}_i$. Based on this, operations such as softmax normalization, vector weighted summation, vector squashing, and coupling coefficient updating are performed, where the intermediate variable b is initialized to 0. Through multiple iterations, the final higher-level capsules are obtained. It can be observed that the larger the inner product between the intermediate vector a^i and the higher-level feature $\boldsymbol{U}_i$, the greater the updated coupling coefficient becomes, enabling the lower-level capsule to transmit more informative features to the higher-level capsule.

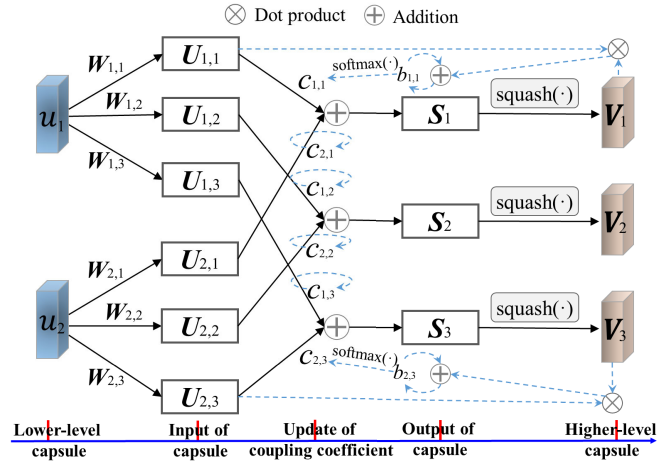


Fig. 4. Schematic of dynamic routing algorithm.

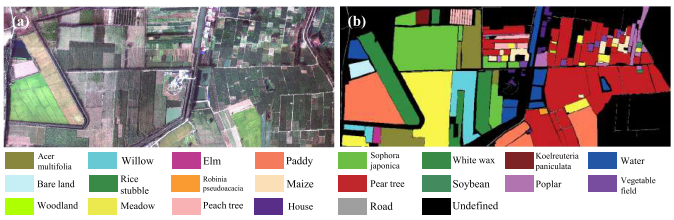


Fig. 5. Hyperspectral dataset of MV. (a) False-color image. (b) Label.

D. Decoder and Loss Function

The decoder is primarily implemented using fully connected layers, with the dimensional transitions of the layers being [512]->[1024]->[patch size × patch size ×C/2]. The specific computation is as follows:

$$\boldsymbol{O}_r = \text{Reshape}\,(\mathbf{W}\boldsymbol{x}) \tag{13}$$

where $\boldsymbol{x}$ denotes the input vector, $\boldsymbol{W}$ is the weight matrix, Reshape(·) represents the reshaping operation, and $\boldsymbol{O}_r$ denotes the reconstructed data.

Unlike the traditional cross-entropy loss function, the loss function of AgriCapsNet consists of both a margin loss and a reconstruction loss. The reconstruction loss serves as a regularization term, computed as the sum of squared differences between the decoder's reconstructed output and the original input. The overall loss function is defined as follows:

$$\begin{cases} L_c = T_c\ max\left(0, m^+ - \|\boldsymbol{v}_c\|\right)^2 \\ \qquad\quad +\lambda\left(1-T_c\right)\ max\left(0, \|\boldsymbol{v}_c\| - m^-\right)^2 \\ L_{\text{total}} = \sum_c L_c + \delta\left\|\boldsymbol{I} - \boldsymbol{O}_r\right\|^2. \end{cases} \tag{14}$$

Here, L_c represents the margin loss for a single capsule neuron, and c denotes the class label. T_c is an indicator function that equals 1 if the class is c, and 0 otherwise. The m^+ and m^- represent the upper and lower margin thresholds (with $m^+ = 0.9$, $m^- = 0.1$), respectively. The former penalizes cases where

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TABLE I
CLASSES AND SAMPLE INFORMATION OF THE MV DATASET

Dataset	Code	Class	Train (1%)	Test	Total
MV	1	Acer multifolia	2256	223391	225647
	2	Willow	1808	178958	180766
	3	Elm	154	15199	15353
	4	Paddy	4521	447623	452144
	5	Sophora japonica	4756	470835	475591
	6	White wax	1693	167649	169342
	7	Koelreuteria paniculata	233	23071	23304
	8	Water	1656	163991	165647
	9	Bare land	384	38025	38409
	10	Rice stubble	1938	191892	193830
	11	Robinia pseudoacacia	56	5556	5612
	12	Maize	592	58573	59165
	13	Pear tree	10264	1016185	1026449
	14	Soybean	72	7079	7151
	15	Poplar	910	90137	91047
	16	Vegetable field	291	28857	29148
	17	Woodland	15	1481	1496
	18	Meadow	4218	417572	421790
	19	Peach tree	655	64859	65514
	20	House	296	29320	29616
	21	Road	422	41786	42208
Total			37192	3682037	3719229

the classifier incorrectly predicts the presence of a class, while the latter penalizes cases where the classifier fails to predict a class that actually exists. λ is a sparsity coefficient used to balance the loss, set to 0.5. L_{total} denotes the total loss, $\boldsymbol{I}$ represents the original input, $\boldsymbol{O}_r$ is the reconstructed output from the decoder, and δ is the weighting factor, set to 2×10^{-5}.

III. EXPERIMENTS AND ANALYSIS

A. Dataset

The hyperspectral images used in this study include the Matiwan Village (MV) [50], the WHU-Hi-HongHu (WHH) [51], and the Zhang-Jia-Gang (ZJG). The first two are publicly available datasets from the remote sensing community, while the latter is a hyperspectral dataset constructed independently for this work (https://doi.org/10.5281/zenodo.18742758). All three datasets encompass complex agricultural landscapes, such as fragmented cropland parcels, diverse crop types, and various other land cover classes.

The MV dataset was acquired by an airborne multimodular imaging spectrometer (AMMIS) [52] at Matiwan, Xiong'an, Hebei, in October 2017. The image size is 1580×3750 pixels with a spatial resolution of 0.5 m and contains 256 spectral bands. The dataset includes 21 land cover classes, among which 14 classes correspond to seedlings, cash crops, and food crops. During model training, 1% of samples from each class were selected via stratified sampling. The class names and detailed sample information are listed in Table I, and the corresponding illustrations are shown in Fig. 5.

The WHH dataset was acquired using the Headwall Nano-Hyperspec sensor in Honghu, Hubei, in December 2017. The image has a size of 940×475 pixels, a spatial resolution of 0.043 m, and contains 270 spectral bands. The dataset consists of 22 land cover classes, including 17 types of cash crops. 100 samples from each class were randomly selected for model training. The class names and detailed sample information are listed in Table II, and the corresponding illustration is shown in Fig. 6.

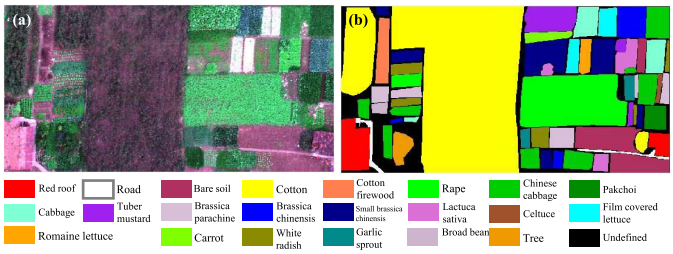


Fig. 6. Hyperspectral dataset of WHH. (a) False-color image. (b) Label.

TABLE II
CLASSES AND SAMPLE INFORMATION OF THE WHH DATASET

Dataset	Code	Class	Train	Test	Total
WHH	1	Red roof	100	13941	14041
	2	Road	100	3412	3512
	3	Bare soil	100	21721	21821
	4	Cotton	100	163185	163285
	5	Cotton firewood	100	6118	6218
	6	Rape	100	44457	44557
	7	Chinese cabbage	100	24003	24103
	8	Pakchoi	100	3954	4054
	9	Cabbage	100	10719	10819
	10	Tuber mustard	100	12294	12394
	11	Brassica parachinensis	100	10915	11015
	12	Brassica chinensis	100	8854	8954
	13	Small brassica chinensis	100	22407	22507
	14	Lactuca sativa	100	7256	7356
	15	Celtuce	100	902	1002
	16	Film covered lettuce	100	7162	7262
	17	Romaine lettuce	100	2910	3010
	18	Carrot	100	3117	3217
	19	White radish	100	8612	8712
	20	Garlic sprout	100	3386	3486
	21	Broad bean	100	1228	1328
	22	Tree	100	3940	4040
Total			2200	384493	386693

The original imagery of the ZJG dataset was provided by the Land Satellite Remote Sensing Application Center, Ministry of Natural Resources of China (LASAC), and was acquired in

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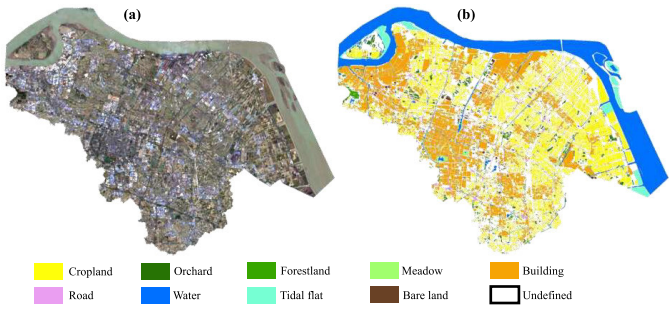


Fig. 7. Hyperspectral dataset of ZJG. (a) False-color image. (b) Label.

TABLE III
CLASSES AND SAMPLE INFORMATION OF THE ZJG DATASET

Dataset	Code	Class	Train (2%)	Test	Total
ZJG	1	Cropland	3451	169076	172527
	2	Orchard	245	11999	12244
	3	Forestland	212	10402	10614
	4	Meadow	161	7885	8046
	5	Building	4473	219156	223629
	6	Road	570	27921	28491
	7	Water	3458	169431	172889
	8	Tidal flat	371	18180	18551
	9	Bare land	80	3920	4000
	Total		13020	637971	650991

October 2020 by the Advanced Hyperspectral Imager (AHSI) onboard the ZY1-02D satellite. The dataset covers Zhangjiagang, Jiangsu, and the corresponding ground truth was manually interpreted by the research team. The preprocessing workflow included radiometric calibration, atmospheric correction, orthorectification, mosaicking, and cropping. The final image size is 1224×1624 pixels, with a spatial resolution of 30 m and 152 spectral bands after removing the water absorption bands. The dataset contains 9 land cover classes, with 2% of the samples in each class selected using stratified sampling for model training. The class names and detailed sample statistics are provided in Table III, and the corresponding illustration is shown in Fig. 7.

B. Experimental Configuration

All experiments were conducted on a desktop workstation equipped with 128 GB of RAM, a 10-core Intel Core i9-9900X CPU, and an NVIDIA GeForce RTX 2080Ti GPU. The programs were implemented in Python. The evaluation results (mean $\pm$ standard deviation) were obtained from 10 independent runs of the model on each dataset. The deep learning frameworks used include PyTorch and Keras [53].

The evaluation metrics used to assess the model include per-class mapping accuracy, overall accuracy (OA), average accuracy (AA), and the Kappa coefficient (κ) [54]. OA represents the probability that the classification results agree with the reference data. AA represents the mean percentage of correctly classified samples per class out of the total number of samples in that class. The advantage of κ lies in its consideration of uncertainty effects on classification results when estimating accuracy. By comparing the classification results with ground truth data, a confusion matrix $\boldsymbol{M}$ is obtained, based on which the evaluation metrics are calculated. The specific formulas are as follows:

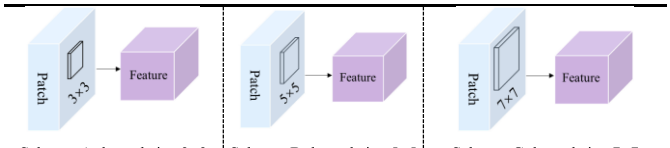
$$OA = \sum_{u=1}^{n} \boldsymbol{M}_{uu} / N \tag{15}$$

$$AA = \left(\sum_{u=1}^{n} \left(\boldsymbol{M}_{uu} \Big/ \sum_{u=1}^{n} \boldsymbol{M}_{uv} \right) \right) \Big/ n \tag{16}$$

$$\kappa = \left(N \left(\sum_{u=1}^{n} \boldsymbol{M}_{uu} \right) - \sum_{u=1}^{n} \left(\sum_{v=1}^{n} \boldsymbol{M}_{uv} \sum_{v=1}^{n} \boldsymbol{M}_{vu} \right) \right) \Big/ \left(N^2 - \sum_{u=1}^{n} \left(\sum_{v=1}^{n} \boldsymbol{M}_{uv} \sum_{v=1}^{n} \boldsymbol{M}_{vu} \right) \right) \tag{17}$$

Here, n denotes the number of classes, N is the total number of test samples, $\boldsymbol{M}_{uu}$ represents the number of samples of class u correctly classified as u, and $\boldsymbol{M}_{uv}$ denotes the number of samples of class v misclassified as u, where $u, v \in [1, n]$.

TABLE IV
CONVOLUTION KERNEL OF LIGHTWEIGHT CONVOLUTION MODULE IN AGRICAPSNET



Scheme-A: kernel size 3×3 | Scheme-B: kernel size 5×5 | Scheme-C: kernel size 7×7

C. Parameter Setting

1) AgriCapsNet Parameter Configuration: Parameter settings of AgriCapsNet mainly consist of two parts: general parameters during training and structural parameters of the network itself.

During model training, the Adam optimizer is used with an initial learning rate of 0.001, combined with an exponential decay learning rate strategy, where the new learning rate equals the initial learning rate multiplied by τ raised to the epoch number ($\tau = 0.8$). The total number of training epochs is 20, batch size is 64, and the default patch size is 11×11. The structural parameters of AgriCapsNet mainly focus on the encoder. After extensive experiments, the intrinsic feature dimension in the lightweight convolution module is set to 128, and the final feature dimension of lightweight convolution module is set to 256. Practical experience indicates [36], [37] that the capsule neuron size in DigitCaps is usually set to twice that in PrimaryCaps. In AgriCapsNet, there are 32 convolutional units in PrimaryCaps with a capsule neuron size of 8, while the capsule neuron size in DigitCaps is set to 16.

Additionally, the impact of the convolution kernel size used in the lightweight convolution module on classification accuracy is investigated, and three schemes were designed while

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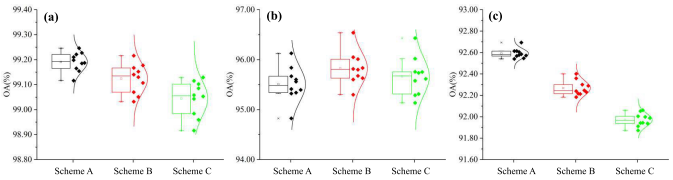


Fig. 8. OA comparison of three schemes on the hyperspectral dataset. (a) MV. (b) WHH. (c) ZJG.

TABLE V
MAIN PARAMETER SETTINGS OF THE INVOLVED DEEP LEARNING NETWORKS

Deep learning networks	Main parameter settings			
	Shared parameters	MV	WHH	ZJG
LSTM [53]	batch size = 64	lr = 0.0001, epochs = 500, grouping strategy = 2, time_step = 3, beta_1 = 0.9, beta_2 = 0.999		
1D-CNN [54]		lr = 0.1, epochs = 2000		
ViT [55]		patch size = 15, lr = 0.0003, epochs = 100, heads = 8		
MSCNN [53]		patch size = 28, lr = 0.0001, epoch = 500, beta_1 = 0.9, beta_2 = 0.999		
3D-CNN [56]		patch size = 11, epochs = 500, 3Dconv1: 32×4×4, 3Dconv2: 32×4×4, 3Dconv3: 32×3×3, stride = 1		
		lr = 0.003	lr = 0.01	lr = 0.003
SSUN [53]		patch size = 28, lr = 0.0001, epochs = 500, grouping strategy = 2, time_step = 3, beta_1 = 0.9, beta_2 = 0.999		
CapsNet-Base [34]		lr = 0.001, epochs = 20, kernel sise = 3		

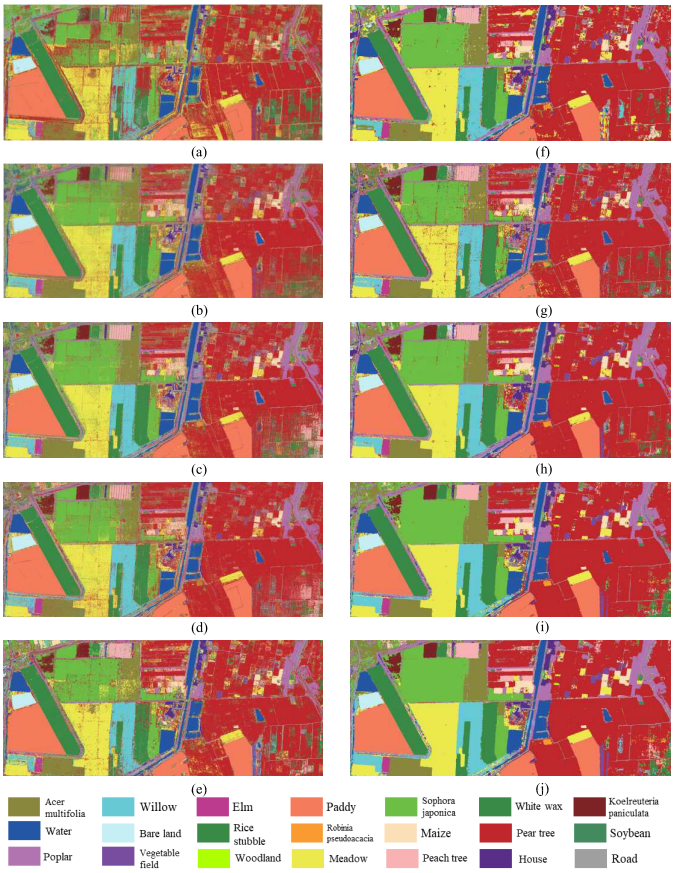


Fig. 9. Classification maps of MV obtained by different models. (a) RF (74.70%). (b) SVM (85.70%). (c) LSTM (90.52%). (d) 1D-CNN(87.11%). (e) ViT (88.44%). (f) MSCNN (96.14%). (g) 3D-CNN (90.89%). (h) SSUN (97.90%). (i) CapsNet-Base (98.29%). (j) AgriCapsNet (99.19%).

keeping the other components of the AgriCapsNet unchanged. Specifically, the convolution kernel sizes were set to 3×3, 5×5, and 7×7, denoted as schemes A, B, and C, respectively. The calculation procedure is illustrated in Table IV.

Each scheme was independently tested ten times on the MV, WHH, and ZJG datasets. The corresponding OA values were calculated, and the results are visualized in Fig. 8, which presents both the boxplot statistics and the normal distributions of OA. It can be observed that the optimal schemes for the MV, WHH, and ZJG datasets are A, B, and A, respectively. Therefore, when configuring the lightweight convolution module for different hyperspectral images, the kernel size should be selected according to the optimal scheme identified for each specific dataset.

2) Parameter Settings of Comparative Models: The models involved in the comparison include SVM [55], RF [56], LSTM [57], 1D-CNN [58], Vision Transformer (ViT) [59], MSCNN [57], 3D-CNN [60], SSUN [57], and CapsNet-Base [38]. To maximize the classification performance of each model, the relevant parameters were tuned. For SVM, the radial basis function (RBF) kernel was employed, and the regularization parameter C and kernel coefficient γ were determined via grid search within the ranges C $\in$ {20, 40, 60, 80, 100}, $\gamma \in$ {0.001, 0.005, 0.01, 0.05, 0.1, 0.125}. For RF, grid search was also applied to key parameters, namely, the number of estimators (NEs) and the maximum depth of estimators (MDEs), with search ranges NE $\in$ {100, 200, 300, 400, 500}, MDE $\in$ {10, 25, 50, 75, 100}. For deep learning networks, parameter settings were initialized according to the original literature and further optimized to achieve the best classification performance across different hyperspectral datasets. The main parameter configurations are summarized in Table V.

D. Comparison With the State of the Arts

Different methods were evaluated for classification performance on the three hyperspectral datasets from MV, WHH, and ZJG. The corresponding classification accuracy metrics were computed, with detailed quantitative evaluation results presented in Tables VI–VIII.

The classification accuracy of the MV dataset is presented in Table VI. AgriCapsNet achieves an OA of 99.19% and a κ of 0.9906, which are 0.90%–24.49% and 0.0104–0.2885 higher than those of other methods, respectively. Regarding per-class mapping accuracy, AgriCapsNet attains over 99% in 11 classes, including willow, elm, paddy, sophora japonica, white wax, koelreuteria paniculata, water, bare land, rice stubble, pear tree, and meadow. Additionally, CapsNet-Base, SSUN, and MSCNN also demonstrate strong classification performance. Classification maps obtained by different methods are

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TABLE VI
CLASSIFICATION ACCURACY OF DIFFERENT MODELS ON THE MV DATASET

Class	RF	SVM	LSTM	1D-CNN	ViT	MSCNN	3D-CNN	SSUN	CapsNet-Base	AgriCapsNet
Acer multifolia	60.48±1.00	83.57±5.46	87.80±1.22	84.08±0.36	80.57±2.90	94.29±0.51	82.67±12.40	97.02±0.48	96.90±0.97	98.90±0.26
Willow	63.01±0.79	81.14±18.70	91.67±0.67	90.12±0.32	85.94±2.48	95.69±0.83	90.18±6.46	98.83±0.41	99.30±0.99	99.70±0.11
Elm	55.79±2.04	80.75±10.48	85.86±2.05	81.08±2.28	83.00±6.19	95.49±0.62	80.74±25.20	98.15±0.54	98.52±0.99	99.17±0.46
Paddy	95.59±0.11	98.53±0.42	98.81±0.08	98.70±0.07	98.56±0.67	99.49±0.09	99.06±0.34	99.67±0.11	99.65±1.00	99.90±0.04
Sophora japonica	63.73±0.47	85.14±4.67	88.66±0.81	84.71±0.32	86.01±2.81	97.02±0.43	87.61±9.30	98.17±0.57	99.42±0.99	99.76±0.05
White wax	62.30±0.65	88.19±5.66	91.78±1.00	89.44±0.34	78.72±6.02	94.02±0.74	86.71±12.39	98.33±0.35	98.95±0.99	99.53±0.15
Koelreuteria paniculata	46.24±1.61	90.23±2.11	89.91±1.68	87.01±1.33	91.50±3.94	96.71±0.74	96.13±5.32	98.73±0.69	99.89±1.00	99.85±0.09
Water	89.87±0.28	95.66±0.80	96.30±0.52	96.20±0.22	96.41±1.60	98.94±0.13	98.26±0.89	99.38±0.21	99.01±0.99	99.70±0.11
Bare land	94.34±0.61	97.16±0.42	97.88±0.34	97.44±0.27	98.76±0.31	98.83±0.38	97.33±1.55	99.39±0.27	99.52±1.00	99.86±0.16
Rice stubble	94.43±0.14	97.37±1.49	98.63±0.10	98.30±0.14	98.33±0.59	99.64±0.07	98.72±1.21	99.82±0.08	99.89±1.00	99.95±0.02
Robinia pseudoacacia	0.00±0.00	49.39±19.21	54.55±2.59	36.45±3.13	26.85±9.19	72.00±6.84	45.15±18.61	87.66±2.98	86.88±0.87	92.80±2.67
Maize	36.47±1.09	70.11±8.51	76.54±2.16	68.33±1.20	76.02±3.47	86.84±1.42	75.75±14.76	92.49±1.55	94.52±0.95	96.49±0.69
Pear tree	82.46±0.32	84.61±10.99	91.49±0.83	87.62±0.15	90.10±1.36	96.77±0.43	93.68±2.15	98.49±0.35	98.82±0.99	99.41±0.13
Soybean	29.32±3.15	51.07±7.28	59.25±3.07	56.63±2.02	52.02±7.84	80.41±4.09	64.96±16.31	87.85±3.10	82.40±0.82	89.82±1.86
Poplar	38.31±1.00	69.82±7.93	77.99±2.57	68.16±1.15	81.96±2.61	91.90±1.24	80.93±7.45	94.39±1.28	94.80±0.95	97.48±0.51
Vegetable field	15.27±0.92	40.27±6.90	46.94±1.66	42.08±1.27	48.51±2.61	69.36±2.05	48.78±5.99	75.82±4.31	81.13±0.81	90.09±1.66
Woodland	0.00±0.00	12.70±7.85	15.06±4.15	5.58±1.41	9.89±3.91	41.43±4.80	15.84±6.45	50.03±6.34	57.00±0.57	76.29±5.44
Meadow	77.17±0.47	80.45±11.50	88.00±1.36	82.02±0.34	86.91±1.87	95.76±0.51	89.89±4.74	97.64±0.44	98.18±0.98	99.19±0.08
Peach tree	24.93±0.84	71.87±5.02	75.43±1.83	65.47±0.67	75.44±4.82	91.35±0.80	78.34±13.22	93.66±2.28	95.37±0.95	98.08±0.49
House	73.05±1.50	82.09±1.24	83.48±1.47	82.04±0.83	86.25±2.58	91.29±1.06	87.20±2.70	94.21±1.64	92.63±0.93	96.73±0.98
Road	63.68±0.89	75.00±6.31	82.01±3.09	78.52±1.32	86.32±4.08	88.70±1.04	83.44±4.31	91.45±1.47	86.22±0.86	92.02±0.88
OA(%)	74.70±0.13	85.70±5.69	90.52±0.12	87.11±0.07	88.44±0.99	96.14±0.20	90.89±4.38	97.90±0.17	98.29±0.10	99.19±0.04
AA(%)	55.54±0.15	75.48±4.30	79.91±0.39	75.24±0.32	77.05±1.34	89.33±0.45	80.07±0.76	92.91±0.60	93.29±0.66	96.41±0.40
$\kappa\times100$	70.21±0.15	83.59±6.28	89.03±0.13	85.09±0.08	86.62±1.14	95.53±0.23	89.40±5.16	97.58±0.20	98.02±0.11	99.06±0.04

shown in Fig. 9, where AgriCapsNet produces smoother and more complete results with significantly reduced salt-and-pepper noise.

Table VII presents the classification accuracy of the WHH dataset. AgriCapsNet achieves the highest OA of 95.83%, outperforming other models by 1.87%–28.82%. For AA and κ, AgriCapsNet improves over other methods by 2.07%–31.98% and 0.0235–0.3405, respectively. AgriCapsNet attains the highest accuracy in most classes, notably achieving 100% classification accuracy for the broad bean. Fig. 10 shows the classification maps of all models. Focusing on the rape, it is evident that AgriCapsNet provides clearer classification results under small sample conditions, significantly reducing speckle noise interference. Benefiting from multi-scale feature extraction and the integration of multiple network types, MSCNN, SSUN, and CapsNet-Base also demonstrate advantages in land cover classification, with OA values all above 92%.

Table VIII shows the classification accuracy of the ZJG dataset. AgriCapsNet achieves an OA of 92.59%, outperforming other methods by 1.84%–7.09%. Its AA and κ are 73.10% and 0.8987, respectively, representing improvements of 5.61%–23.05% and 0.0261–0.0993 compared to other methods. For all models, the classification accuracies for the classes of forestland, meadow, and road are relatively low. Possible reasons include: 1) forestland and meadow are prone to confusion and have limited sample sizes, causing class imbalance that poses challenges to the classifiers and leads to some misclassifications; and 2) road suffers from severe mixed pixel effects at the 30 m spatial resolution, making classification more difficult. The classification maps of all methods are shown in Fig. 11, where AgriCapsNet produces more accurate classification results in cropland, building, water, and tidal flat.

IV. DISCUSSION

A. Effect of Reconstruction Loss on AgriCapsNet

To investigate the effect of the reconstruction loss, which serves as a regularization term in the loss function, on the classification performance of AgriCapsNet, the weight factor δ was varied across nine settings $\{10^{-5}, 2\times10^{-5}, 5\times10^{-5}, 10^{-4}, 2\times10^{-4}, 5\times10^{-4}, 10^{-3}, 2\times10^{-3}, 5\times10^{-3}\}$, and the proposed model was evaluated on the three hyperspectral

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TABLE VII
CLASSIFICATION ACCURACY OF DIFFERENT MODELS ON THE WHH DATASET

Class	RF	SVM	LSTM	1D-CNN	ViT	MSCNN	3D-CNN	SSUN	CapsNet-Base	AgriCapsNet
Red roof	80.54±1.74	90.24±0.47	89.02±1.41	89.86±1.19	95.95±0.62	95.69±1.08	93.52±0.88	96.77±1.49	98.11±0.92	98.14±0.90
Road	81.17±2.15	81.49±0.94	84.14±1.79	86.94±1.75	93.42±0.55	95.19±1.64	94.77±1.50	94.87±1.74	98.10±0.80	98.37±0.64
Bare soil	68.43±1.49	70.79±1.40	70.13±4.08	71.39±2.01	90.90±0.72	93.13±2.43	87.77±2.07	92.58±1.59	90.34±2.74	93.22±2.03
Cotton	72.11±1.55	80.09±0.75	79.71±1.93	80.81±2.01	94.78±0.72	97.67±0.58	94.07±1.36	98.28±0.64	96.65±0.89	96.99±0.62
Cotton firewood	67.72±2.20	74.93±1.28	73.07±3.35	73.18±2.15	95.88±1.22	95.51±2.30	88.81±1.73	98.36±0.76	97.10±1.11	98.01±0.71
Rape	74.58±2.22	79.67±0.62	81.33±2.02	82.27±0.91	86.52±1.23	91.39±1.05	93.66±0.76	92.09±1.41	93.97±0.90	96.65±1.12
Chinese cabbage	46.77±3.18	59.10±1.93	59.65±2.84	59.49±2.68	64.39±1.83	69.54±2.40	70.72±3.42	75.37±3.16	85.24±3.60	91.81±2.12
Pakchoi	26.81±2.41	41.34±1.60	45.10±3.44	46.04±1.39	69.12±2.75	89.87±3.03	75.22±5.45	91.27±2.98	95.01±2.69	98.47±1.09
Cabbage	87.38±1.46	91.09±0.55	92.83±0.87	92.51±0.72	93.33±2.40	94.43±1.27	94.49±1.02	96.45±1.01	98.39±0.84	98.71±0.69
Tuber mustard	34.91±2.63	54.76±3.09	57.96±1.56	55.21±0.89	77.65±4.11	88.95±2.17	84.61±7.37	90.12±2.74	91.89±2.36	93.63±1.61
Brassica parachinensis	31.32±3.81	42.03±1.51	47.27±2.88	46.12±1.97	62.96±3.20	79.87±1.96	77.53±3.48	79.03±4.61	89.68±2.87	93.95±1.63
Brassica chinensis	58.61±2.49	55.18±3.61	57.53±2.25	57.42±1.94	80.86±2.56	84.92±3.22	78.59±2.58	85.98±2.84	79.48±7.18	93.37±2.29
Small brassica chinensis	54.49±1.76	50.73±1.58	52.93±3.01	49.47±1.75	70.02±1.93	78.54±2.07	74.91±3.35	81.85±2.25	83.23±1.80	86.95±1.15
Lactuca sativa	55.01±2.14	69.02±1.08	71.25±2.33	69.04±2.78	81.13±2.18	92.90±1.94	90.44±1.70	94.30±1.27	96.14±1.53	97.80±0.65
Celtuce	74.07±1.74	86.71±0.98	87.90±1.82	87.05±1.75	92.53±2.10	98.57±0.63	96.71±1.08	98.17±0.90	99.40±0.72	99.78±0.47
Film covered lettuce	77.10±2.32	78.12±1.18	81.31±2.80	78.72±2.27	95.04±1.26	93.30±2.19	96.04±0.75	93.93±1.39	98.49±0.59	98.78±0.49
Romaine lettuce	65.87±2.60	76.81±0.78	77.92±2.45	77.08±2.33	96.23±1.34	97.00±1.79	94.44±3.51	97.21±1.91	98.97±0.72	99.70±0.25
Carrot	70.06±2.61	74.37±1.30	72.99±2.11	73.26±1.77	89.44±2.64	97.03±1.02	93.70±1.81	96.39±1.32	99.02±0.43	99.40±0.61
White radish	73.59±1.90	77.09±1.68	78.42±2.02	77.61±1.92	84.77±3.07	86.32±2.16	88.15±2.04	91.94±0.84	95.94±1.12	96.60±0.63
Garlic sprout	76.08±1.66	83.47±1.33	81.12±1.28	80.14±1.66	94.45±1.32	95.88±1.96	94.71±0.35	96.05±0.87	99.07±0.53	99.29±0.46
Broad bean	74.58±2.63	68.84±2.01	66.64±4.98	63.59±2.96	97.50±1.11	99.15±0.43	97.01±1.78	98.16±1.30	100.00±0.00	100.00±0.00
Tree	74.15±2.80	73.75±1.93	73.43±3.28	72.43±3.26	90.59±1.63	98.82±0.68	94.97±2.56	98.18±0.49	99.11±1.08	99.34±0.67
OA(%)	67.01±0.88	73.80±0.41	74.40±0.79	74.62±0.99	87.69±0.60	92.01±0.38	89.55±0.59	93.21±0.44	93.96±0.58	95.83±0.37
AA(%)	64.79±0.37	70.89±0.18	71.89±0.43	71.35±0.36	86.25±0.89	91.53±0.41	88.86±0.37	92.61±0.47	94.70±0.41	96.77±0.24
κ×100	60.70±0.92	68.28±0.44	69.04±0.83	69.20±1.07	84.62±0.74	89.95±0.47	86.95±0.70	91.45±0.54	92.40±0.71	94.75±0.46

TABLE VIII
CLASSIFICATION ACCURACY OF DIFFERENT MODELS ON THE ZJG DATASET

Class	RF	SVM	LSTM	1D-CNN	ViT	MSCNN	3D-CNN	SSUN	CapsNet-Base	AgriCapsNet
Cropland	92.64±0.28	92.40±0.20	93.75±1.79	92.31±0.81	92.86±0.69	91.95±0.66	93.42±0.70	95.22±0.80	97.29±0.24	97.27±0.18
Orchard	33.91±1.86	42.10±0.68	42.67±3.79	40.14±3.61	43.06±3.19	48.82±4.00	48.19±6.37	55.69±1.80	56.35±2.54	67.62±1.14
Forestland	29.82±1.99	34.96±1.55	34.88±3.63	37.08±4.29	36.41±2.00	35.50±3.42	38.18±6.16	44.59±2.35	38.43±3.02	48.56±1.85
Meadow	4.61±1.55	17.69±1.11	14.78±2.94	22.19±3.68	21.71±1.75	29.78±3.51	23.84±3.46	33.85±3.16	21.97±3.03	33.84±2.40
Building	91.57±0.44	88.46±0.17	90.99±1.33	88.80±1.35	90.55±1.65	89.64±0.95	91.97±1.06	92.41±0.56	95.27±0.22	95.38±0.19
Road	18.71±1.86	36.49±1.37	39.44±8.08	36.76±5.44	28.10±1.34	30.52±2.82	39.24±3.37	39.54±1.87	34.97±2.37	53.00±1.01
Water	96.07±0.17	96.92±0.15	97.68±0.26	97.07±0.20	96.33±0.16	95.97±0.24	97.12±0.33	97.32±0.20	97.35±0.24	98.27±0.09
Tidal flat	65.54±1.23	82.92±0.78	85.34±2.12	81.89±1.11	82.51±1.49	87.91±2.27	83.25±2.96	91.54±0.87	92.37±0.90	95.79±0.45
Bare land	17.53±2.97	44.41±2.29	54.20±3.94	50.41±3.08	33.92±3.92	47.21±3.70	43.79±7.00	57.21±3.72	42.57±4.19	68.17±4.25
OA(%)	85.50±0.09	86.43±0.11	88.09±0.27	86.63±0.22	86.76±0.51	86.65±0.23	88.33±0.27	89.71±0.19	90.75±0.24	92.59±0.04
AA(%)	50.05±0.58	59.60±0.43	61.53±1.08	60.74±0.72	58.38±0.55	61.92±1.30	62.11±1.47	67.49±0.75	64.06±1.10	73.10±0.47
κ×100	79.94±0.14	81.52±0.15	83.74±0.34	81.82±0.29	81.89±0.66	81.76±0.30	84.07±0.36	85.95±0.25	87.26±0.33	89.87±0.06

datasets. Fig. 12 presents the corresponding OA and κ. As shown in Fig. 12, variations in δ have a considerable impact on classification outcomes. Specifically, the OA fluctuations for the MV, WHH, and ZJG datasets are 0.86%, 0.93%, and 2.08%, respectively, while the κ fluctuations are 0.99%, 1.16%, and 2.94%, respectively. Overall, the classification performance of AgriCapsNet remains stable across the three datasets when $\delta \leq 5 \times 10^{-4}$. Therefore, it is recommended that δ

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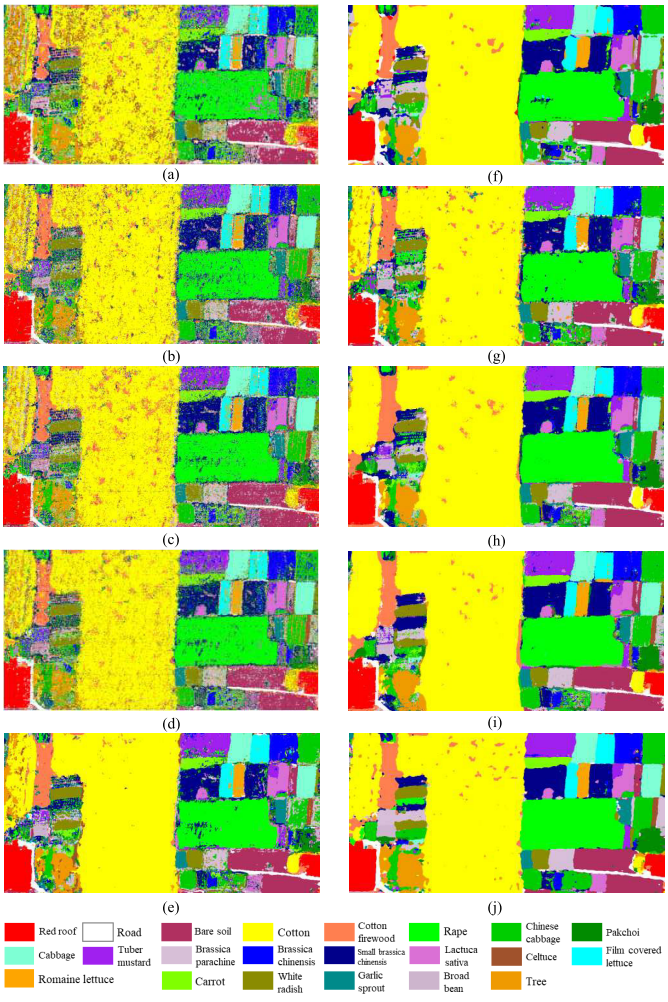


Fig. 10. Classification maps of WHH obtained by different models. (a) RF (67.01%). (b) SVM (73.80%). (c) LSTM (74.40%). (d) 1D-CNN (74.62%). (e) ViT (87.69%). (f) MSCNN (92.01%). (g) 3D-CNN (89.55%). (h) SSUN (93.21%). (i) CapsNet-Base (93.96%). (j) AgriCapsNet (95.83%).

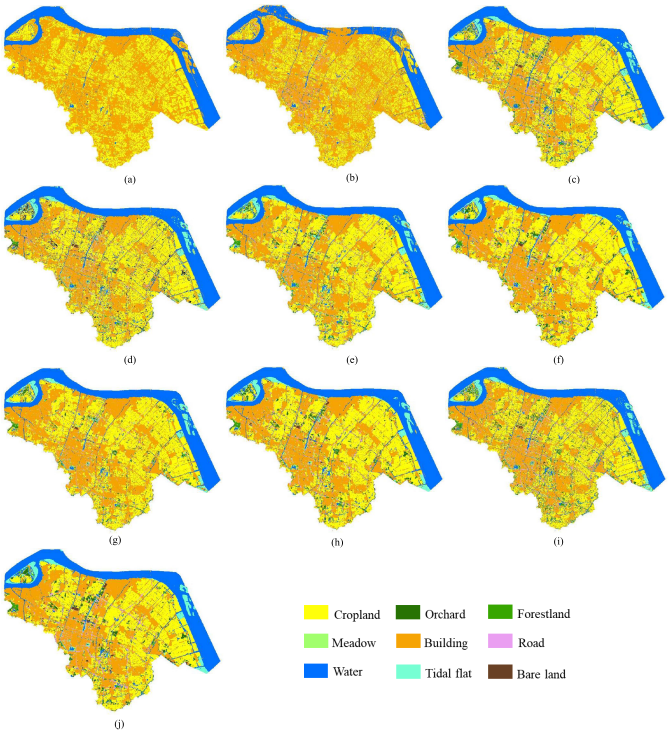


Fig. 11. Classification maps of ZJG obtained by different models. (a) RF (85.50%). (b) SVM (86.43%). (c) LSTM (88.09%). (d) 1D-CNN (86.63%). (e) ViT (86.76%). (f) MSCNN (86.65%). (g) 3D-CNN (88.33%). (h) SSUN (89.71%). (i) CapsNet-Base (90.75%). (j) AgriCapsNet (92.59%).

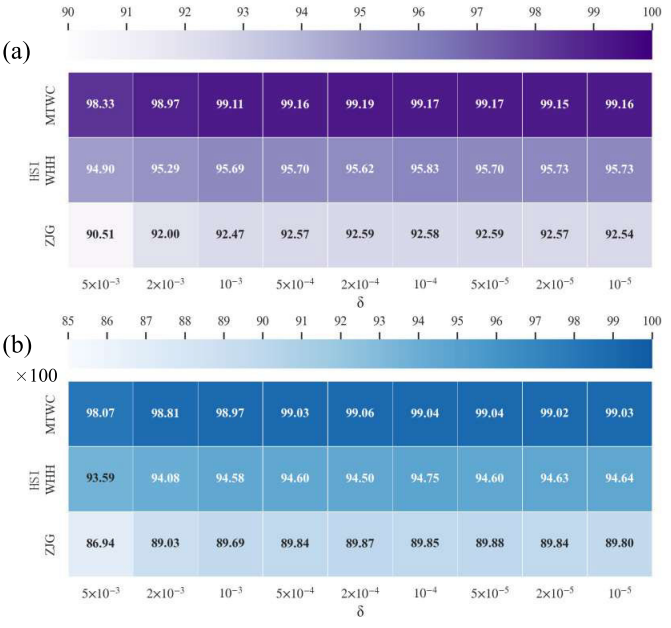


Fig. 12. Impact of different weight factor δ on the classification performance of AgriCapsNet. (a) OA. (b) κ.

be set no higher than 5×10^{-4}. For different classification tasks and data sources, the value of δ should be adjusted according to the specific application to achieve optimal AgriCapsNet performance. In this study, $\delta = 2 \times 10^{-4}$ is recommended for the MV dataset (spatial resolution: 0.5 m) and the WHH dataset (spatial resolution: 0.043 m), while $\delta = 10^{-4}$ is recommended for the ZJG dataset (spatial resolution: 30 m). These recommended values can serve as References for AgriCapsNet-based classification of HSI with similar spatial resolutions.

B. Impact of Patch Size on the Classification Performance of AgriCapsNet

Patch size often significantly affects classification performance, especially for networks with convolutional inputs. Since AgriCapsNet employs a lightweight convolution module at the input stage, it is necessary to investigate its sensitivity to different patch sizes. For the three hyperspectral datasets, training and testing samples were allocated as detailed in Tables I–III, and the input patch sizes were set to 9×9, 11×11, 13×13, 15×15, 17×17, and 19×19. The classification performance of AgriCapsNet under these settings was evaluated, and the corresponding OA and κ values are presented in Fig. 13.

Experiments on various hyperspectral datasets indicate that AgriCapsNet's classification accuracy does not always increase with larger patch sizes. Instead, it is closely related to the spatial resolution of the imagery. For the high spatial

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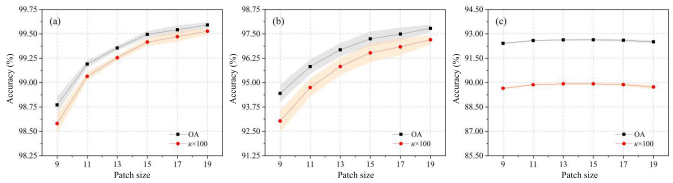


Fig. 13. Classification accuracy of AgriCapsNet on hyperspectral images with different patch sizes. (a) MV. (b) WHH. (c) ZJG.

TABLE IX
ABLATION TEST RESULTS OF AGRICAPSNET ON DIFFERENT HSIS

Scheme	Spatial attention	Channel attention	LCM	Capsule layer	HSI	OA (%)	κ×100
scheme-1	√	√	√	√	MV	99.19	99.06
					WHH	95.83	94.75
					ZJG	92.59	89.87
scheme-2	×	√	√	√	MV	99.09	98.95
					WHH	95.63	94.50
					ZJG	92.46	89.69
scheme-3	√	×	√	√	MV	99.15	99.01
					WHH	95.71	94.61
					ZJG	92.56	89.82
scheme-4	√	√	×	√	MV	99.14	99.00
					WHH	94.88	93.57
					ZJG	92.53	89.79
scheme-5	×	×	×	√	MV	98.29	98.02
					WHH	93.96	92.40
					ZJG	90.75	87.26

LCM: Lightweight convolution Module.

resolution datasets MV and WHH, AgriCapsNet's classification accuracy consistently improves with increasing patch size. Conversely, for the medium-resolution ZJG dataset, the classification accuracy initially rises but subsequently declines as patch size grows. Notably, when the patch size ranges from 11×11 to 17×17 for ZJG, the OA remains above 92.59%, and κ exceeds 0.89, demonstrating AgriCapsNet's robust feature extraction even with relatively small patches. Furthermore, classification accuracy of AgriCapsNet decreases when using small patches (e.g., 9×9) or large patches (e.g., 19×19). This can be attributed to two main factors: 1) small patches provide limited spatial context, restricting the neighborhood information the model can leverage, especially for high-resolution imagery, leading to poorer classification; and 2) each pixel covers a larger area for lower spatial resolution images, and large patches in such images may introduce significant mixed pixel effects, hindering the classifier's ability to accurately identify the target.

Therefore, when applying AgriCapsNet to classification tasks, it is recommended to appropriately increase the patch size for high-resolution hyperspectral images to expand the spatial context and enhance feature representation, while keeping patch sizes moderate for medium- or low-resolution images to avoid accuracy degradation.

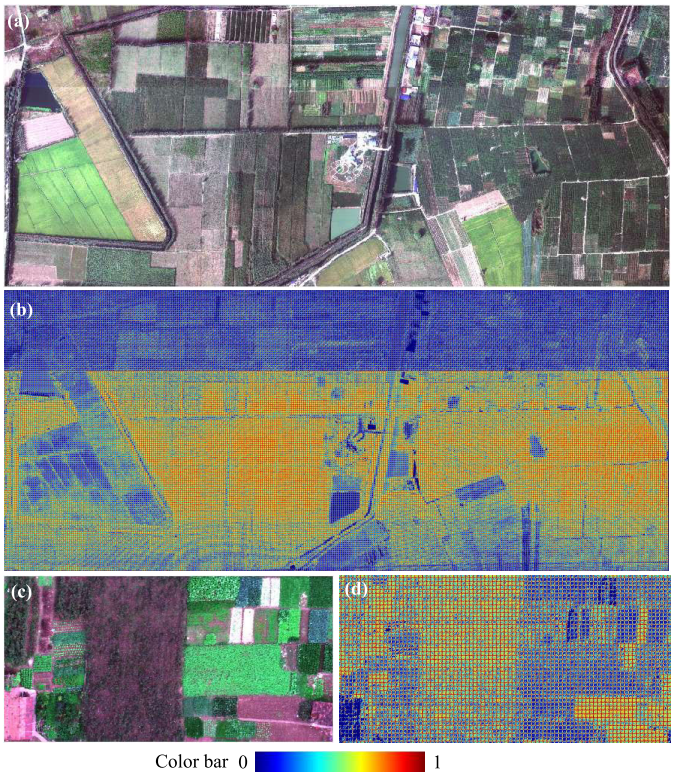


Fig. 14. Visualization of spatial attention features generated by AgriCapsNet for the MV and WHH datasets. For each dataset, the false-color composite image and the corresponding spatial attention map are presented. The attention maps are reconstructed by stitching image patches of size 11×11, resulting in a visible checkerboard artifact. (a) False-color image of MV. (b) Spatial attention map of MV. (c) False-color image of WHH. (d) Spatial attention map of WHH.

C. Ablation Study

The AgriCapsNet comprises four key components, including spatial attention, channel attention, a lightweight convolution module, and a capsule layer. To investigate the contribution of each component to the model's classification performance, five ablation schemes (scheme-1 to scheme-5) were designed, and their classification accuracies (OA and κ) were evaluated on three hyperspectral datasets, as summarized in Table IX. Based on the classification results, the components can be ranked in terms of contribution as follows: spatial attention > lightweight convolution module > channel attention. Although scheme-5 achieves relatively lower classification accuracy on all three datasets, it still delivers competitive results, thereby ensuring a solid baseline performance for AgriCapsNet. Overall, the ablation results indicate that scheme-1 (i.e., AgriCapsNet) remains the optimal configuration, indirectly validating the effectiveness of the proposed architecture.

D. Feature Representation Capability of AgriCapsNet

To intuitively illustrate AgriCapsNet's capability in representing land cover features, the MV and WHH datasets are

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taken as examples, and the features generated by the spatial attention module are visualized, as shown in Fig. 14. These visualizations include the false-color composite images and the corresponding spatial attention maps. Since each attention map for the entire image is obtained by stitching together individual image patches (size: 11 × 11), a checkerboard artifact is visually noticeable.

From Fig. 14, it can be observed that the key land cover features in the imagery are effectively activated, resulting in spatially varying feature weights for the input data. Specifically, based on the model's learned feature importance, higher weights (approaching 1) are assigned to critical regions, while lower weights (approaching 0) are assigned to less relevant areas, thereby enhancing the model's capability to filter the input features. The spatial attention maps clearly delineate the complete contours of objects, with the values in areas covered by targets exhibiting a distinct hierarchical distribution, which facilitates the efficient extraction of class-specific features. Notably, the MV dataset is composed of two stitched images. Due to color differences on either side of the seam line, the activation patterns in the spatial attention maps differ across the boundary. However, the classification results show that the image stitching has a negligible impact on the final classification accuracy.

E. Model Running Efficiency

The runtime efficiency and complexity of a model are generally influenced by various factors, including network type, model architecture, input data, and batch size. After a model reaches a certain level of classification performance, its operational efficiency becomes a critical consideration. In large-scale deep learning applications, model training and tuning are essential processes. To evaluate model complexity, the training time, number of parameters, and FLOPs of different models were summarized in Table X. Due to differences in parameter settings, the parameter counts and FLOPs vary across models corresponding to different experimental datasets. Since different deep learning models require varying numbers of epochs to converge, training time is reported in seconds per epoch (s/epoch), averaged over ten epochs.

When the data volume increases, networks with relatively longer training times include 3D-CNN, ViT, and 1D-CNN. Taking the MV dataset as an example, which contains 37 192 training samples, the training time per epoch for these three models is 150.97, 113.25, and 52.03 s, respectively, which is several times longer than that of other models. Possible reasons include the following:

1) Although 3D-CNN and ViT have relatively small parameter counts, they use 3-D convolution kernels and numerous multi-head attention modules, respectively, significantly increasing computational cost and complexity. Their FLOPs at a batch size of 1 are 1911.50 and 69.04 M, respectively, resulting in higher time consumption.
2) The 1D-CNN has relatively low parameter count and FLOPs, but pointwise and grouped convolutions, despite theoretically being more efficient than 3×3 convolutions, suffer from limited parallelism and insufficient optimization in mainstream deep learning libraries. This results in fragmented computations and limited speedup on GPUs [61], [62].

TABLE X
TRAINING TIME (S/EPOCH), PARAMETERS (M), AND FLOPS (M) OF DIFFERENT DEEP LEARNING MODELS

Models	Metrics	MV	WHH	ZJG
LSTM	Training time	10.22	0.68	3.51
	Parameters	0.13	0.13	0.11
	FLOPs	0.34	0.35	0.27
1D-CNN	Training time	52.03	2.49	17.64
	Parameters	0.08	0.08	0.07
	FLOPs	0.21	0.23	0.12
ViT	Training time	113.25	6.15	38.26
	Parameters	2.67	2.69	2.55
	FLOPs	69.04	69.44	66.04
MSCNN	Training time	7.09	0.98	2.91
	Parameters	1.06	1.06	1.06
	FLOPs	2.13	2.13	2.12
3D-CNN	Training time	150.97	8.71	34.73
	Parameters	0.95	0.97	0.85
	FLOPs	1911.50	2047.77	899.27
SSUN	Training time	17.14	1.39	6.44
	Parameters	1.23	1.23	1.20
	FLOPs	2.54	2.55	2.46
CapsNet-Base	Training time	14.85	0.93	3.61
	Parameters	10.42	10.81	7.04
	FLOPs	35.32	36.26	27.14
AgriCapsNet	Training time	30.18	2.23	4.99
	Parameters	10.35	10.88	7.00
	FLOPs	44.21	62.11	37.26

Models with shorter training times include MSCNN, CapsNet-Base, and AgriCapsNet. Among them, MSCNN demonstrates higher overall efficiency across the three hyperspectral datasets. Although AgriCapsNet has a relatively larger number of parameters, its simpler architecture and shallower depth lead to shorter training times, demonstrating potential for large-scale applications. Overall, compared to larger models, AgriCapsNet's parameter count and FLOPs remain within acceptable limits. With the growing adoption of parallel computing and cloud computing technologies, the negative impact of model size is expected to be significantly mitigated.

F. Practical Application Considerations in Agricultural Scenarios

In practical applications, such as large-scale irrigation districts, agricultural landscapes are often characterized by fragmented fields, heterogeneous crop distributions, and varying management practices. These factors introduce significant spectral–spatial complexity and pose challenges for reliable crop classification. The proposed AgriCapsNet demonstrates strong potential adaptability to such scenarios. By leveraging

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its attention mechanism and capsule-based representation, the model can suppress background interference and emphasize informative crop-related features, which is particularly beneficial in irrigated regions and under different farm management scenarios. Moreover, the capsule-based representation facilitates the modeling of local spectral–spatial feature compositions, making the model more robust to mixed pixels and irregular field boundaries commonly observed in irrigation districts. Nevertheless, the practical deployment of AgriCapsNet in large-area agricultural monitoring also faces challenges. The availability of high-quality HSI and representative training samples remains a limiting factor in some regions. In addition, computational efficiency and scalability should be further considered when extending the model to continuous monitoring over extensive irrigation districts.

V. Conclusion

To meet the demand for fine and intelligent classification of complex agricultural landscapes using HSI, this study proposes an AgriCapsNet designed to enhance spatial-spectral feature representation. The proposed network effectively captures both spatial and spectral characteristics of diverse land cover types. Its vector-based neurons enable comprehensive modeling of instantiation parameters for object features, resulting in stronger feature focusing and representation capabilities. AgriCapsNet was evaluated and compared with nine other models on the crop-rich MV, WHH, and ZJG datasets. The results demonstrate that AgriCapsNet consistently outperforms competing methods both qualitatively and quantitatively, achieving OA improvements of 0.90%–24.49%, 1.87%–28.82%, and 1.84%–7.09% across the three datasets, respectively. Moreover, AgriCapsNet exhibits strong robustness against noise and flexibility in adapting to varying pixel neighborhood sizes, highlighting its practical applicability.

References

[1] L. V. Rasmussen et al., "Joint environmental and social benefits from diversified agriculture," *Science*, vol. 384, no. 6691, pp. 87–93, 2024.

[2] Q. Tong, Y. Xue, and L. Zhang, "Progress in hyperspectral remote sensing science and technology in China over the past three decades," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 7, no. 1, pp. 70–91, Jan. 2014.

[3] B. Kumar, O. Dikshit, A. Gupta, and M. K. Singh, "Feature extraction for hyperspectral image classification: A review," *Int. J. Remote Sens.*, vol. 41, no. 16, pp. 6248–6287, 2020.

[4] P. Ghamisi et al., "New frontiers in spectral–spatial hyperspectral image classification: The latest advances based on mathematical morphology, Markov random fields, segmentation, sparse representation, and deep learning," *IEEE Geosci. Remote Sens. Mag.*, vol. 6, no. 3, pp. 10–43, Sep. 2018.

[5] S. Liu, D. Marinelli, L. Bruzzone, and F. Bovolo, "A review of change detection in multitemporal hyperspectral images: Current techniques, applications, and challenges," *IEEE Geosci. Remote Sens. Mag.*, vol. 7, no. 2, pp. 140–158, Jun. 2019.

[6] C. Zhang et al., "Remote sensing for crop mapping: A perspective on current and future crop-specific land cover data products," *Remote Sens. Environ.*, vol. 330, Dec. 2025, Art. no. 114995.

[7] Z. Zhang et al., "UAV hyperspectral remote sensing image classification: A systematic review," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 18, pp. 3099–3124, 2025.

[8] C. Fu, B. Du, and L. Zhang, "ReSC-net: Hyperspectral image classification based on attention-enhanced residual module and spatial-channel attention," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5518615.

[9] S. Jia, G. Tang, J. Zhu, and Q. Li, "A novel ranking-based clustering approach for hyperspectral band selection," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 1, pp. 88–102, Jan. 2016.

[10] A. Taramelli, E. Valentini, C. Innocenti, and S. Cappucci, "FHYL: Field spectral libraries, airborne hyperspectral images and topographic and bathymetric LiDAR data for complex coastal mapping," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2013, pp. 2270–2273.

[11] H. Su, W. Yao, Z. Wu, P. Zheng, and Q. Du, "Kernel low-rank representation with elastic net for China coastal wetland land cover classification using GF-5 hyperspectral imagery," *ISPRS J. Photogramm. Remote Sens.*, vol. 171, pp. 238–252, Jan. 2021.

[12] P. Du et al., "Advances of four machine learning methods for spatial data handling: A review," *J. Geovisualization Spatial Anal.*, vol. 4, no. 1, pp. 1–25, Jun. 2020.

[13] P. Ghamisi, J. Plaza, Y. Chen, J. Li, and A. J. Plaza, "Advanced spectral classifiers for hyperspectral images: A review," *IEEE Geosci. Remote Sens. Mag.*, vol. 5, no. 1, pp. 8–32, Mar. 2017.

[14] Y. Chen et al., "A novel weakly supervised semantic segmentation framework to improve the resolution of land cover product," *ISPRS J. Photogramm. Remote Sens.*, vol. 196, pp. 73–92, Feb. 2023.

[15] W. Zhang et al., "A novel knowledge-driven automated solution for high-resolution cropland extraction by cross-scale sample transfer," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023.

[16] N. R. Rao, "Development of a crop-specific spectral library and discrimination of various agricultural crop varieties using hyperspectral imagery," *Int. J. Remote Sens.*, vol. 29, no. 1, pp. 131–144, 2008.

[17] J. A. Benediktsson, M. Pesaresi, and K. Arnason, "Classification and feature extraction for remote sensing images from urban areas based on morphological transformations," *IEEE Trans. Geosci. Remote Sens.*, vol. 41, no. 9, pp. 1940–1949, Sep. 2003.

[18] M. Dalla Mura, A. Villa, J. A. Benediktsson, J. Chanussot, and L. Bruzzone, "Classification of hyperspectral images by using extended morphological attribute profiles and independent component analysis," *IEEE Geosci. Remote Sens. Lett.*, vol. 8, no. 3, pp. 542–546, May 2011.

[19] S. Jia, L. Shen, and Q. Li, "Gabor feature-based collaborative representation for hyperspectral imagery classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 2, pp. 1118–1129, Feb. 2015.

[20] W. Zhang et al., "An improved feature set for hyperspectral image classification: Harmonic analysis optimized by multiscale guided filter," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, pp. 3903–3916, 2020.

[21] F. Li, J. Wang, R. Lan, Z. Liu, and X. Luo, "Hyperspectral image classification using multi-feature fusion," *Opt. Laser Technol.*, vol. 110, pp. 176–183, Feb. 2019.

[22] C. Chen, W. Li, H. Su, and K. Liu, "Spectral–spatial classification of hyperspectral image based on kernel extreme learning machine," *Remote Sens.*, vol. 6, no. 6, pp. 5795–5814, Jun. 2014.

[23] J. Xia, P. Du, X. He, and J. Chanussot, "Hyperspectral remote sensing image classification based on rotation forest," *IEEE Geosci. Remote Sens. Lett.*, vol. 11, no. 1, pp. 239–243, Jan. 2014.

[24] L. Wei, M. Yu, Y. Zhong, Z. Yuan, and C. Huang, "Hyperspectral image classification method based on space-spectral fusion conditional random field," *Acta Geodaetica et Cartographica Sinica*, vol. 49, no. 3, pp. 343–354, 2020.

[25] Q. Zhu et al., "Knowledge-guided land pattern depiction for urban land use mapping: A case study of Chinese cities," *Remote Sens. Environ.*, vol. 272, Apr. 2022, Art. no. 112916.

[26] H. Liu, P. Gong, J. Wang, X. Wang, G. Ning, and B. Xu, "Production of global daily seamless data cubes and quantification of global land cover change from 1985 to 2020–iMap world 1.0," *Remote Sens. Environ.*, vol. 258, Jun. 2021, Art. no. 112364.

[27] N. Audebert, B. Le Saux, and S. Lefevre, "Deep learning for classification of hyperspectral data: A comparative review," *IEEE Geosci. Remote Sens. Mag.*, vol. 7, no. 2, pp. 159–173, Jun. 2019.

[28] C. Wang et al., "A review of deep learning used in the hyperspectral image analysis for agriculture," *Artif. Intell. Rev.*, vol. 54, no. 7, pp. 5205–5253, Oct. 2021.

[29] Z. Xue, M. Zhang, Y. Liu, and P. Du, "Attention-based second-order pooling network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 11, pp. 9600–9615, Nov. 2021.

[30] Y. Xu, D. Wang, L. Zhang, and L. Zhang, "Dual selective fusion transformer network for hyperspectral image classification," *Neural Netw.*, vol. 187, Jul. 2025, Art. no. 107311.

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[31] X. Tian, F. Luo, C. Fu, T. Guo, B. Du, and J. Chanussot, "MAGS: Max-gap loss-guided Siamese-reconstruction network for hyperspectral image partial label learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5506813.
[32] F. Luo, Y. Hua, C. Fu, T. Guo, Y. Duan, and G. Shi, "MMD-MLP: LiDAR-guided hyperspectral data classification using local–global directional-MLP with multiresolution multiscale representation," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5508414.
[33] D. Wang et al., "HyperSIGMA: Hyperspectral intelligence comprehension foundation model," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 47, no. 8, pp. 6427–6444, Aug. 2025.
[34] J. Li et al., "HyperFree: A channel-adaptive and tuning-free foundation model for hyperspectral remote sensing imagery," in *Proc. Comput. Vis. Pattern Recognit. Conf.*, 2025, pp. 23048–23058.
[35] G. E. Hinton, A. Krizhevsky, and S. D. Wang, "Transforming auto-encoders," in *Proc. 21st Int. Conf. Artif. Neural Netw. Artif. Neural Netw. Mach. Learn. (ICANN)*, Espoo, Finland, Jun. 2011, pp. 44–51.
[36] S. Sabour, N. Frosst, and G. E. Hinton, "Dynamic routing between capsules," in *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS)*, vol. 30, 2017.
[37] E. Xi, S. Bing, and Y. Jin, "Capsule network performance on complex data," 2017, *arXiv:1712.03480*.
[38] M. E. Paoletti et al., "Capsule networks for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 4, pp. 2145–2160, Apr. 2019.
[39] X. Wang, K. Tan, P. Du, B. Han, and J. Ding, "A capsule-vectored neural network for hyperspectral image classification," *Knowl.-Based Syst.*, vol. 268, May 2023, Art. no. 110482.
[40] H. Wang and L. Wang, "Collaborative active learning based on improved capsule networks for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023.
[41] X. Jiang, W. Liu, Y. Zhang, J. Liu, S. Li, and J. Lin, "Spectral–spatial hyperspectral image classification using dual-channel capsule networks," *IEEE Geosci. Remote Sens. Lett.*, vol. 18, no. 6, pp. 1094–1098, Jun. 2021.
[42] X. Wang, K. Tan, Q. Du, Y. Chen, and P. Du, "Caps-TripleGAN: GAN-assisted CapsNet for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 9, pp. 7232–7245, Sep. 2019.
[43] H.-C. Li, W.-Y. Wang, L. Pan, W. Li, Q. Du, and R. Tao, "Robust capsule network based on maximum correntropy criterion for hyperspectral image classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, pp. 738–751, 2020.
[44] J. Yin, S. Li, H. Zhu, and X. Luo, "Hyperspectral image classification using CapsNet with well-initialized shallow layers," *IEEE Geosci. Remote Sens. Lett.*, vol. 16, no. 7, pp. 1095–1099, Jul. 2019.
[45] Y. Lecun, Y. Bengio, and G. Hinton, "Deep learning," *Nature*, vol. 521, no. 7553, pp. 436–444, 2015.
[46] J. Gu et al., "Recent advances in convolutional neural networks," *Pattern Recognit.*, vol. 77, pp. 354–377, May 2018.
[47] K. Han, Y. Wang, Q. Tian, J. Guo, C. Xu, and C. Xu, "GhostNet: More features from cheap operations," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2020, pp. 1580–1589.
[48] S. Woo, J. Park, J. Y. Lee, and I. S. Kweon, "CBAM: Convolutional block attention module," in *Proc. Eur. Conf. Comput. Vis.*, 2018, pp. 3–19.
[49] Q. Wang, B. Wu, P. Zhu, P. Li, W. Zuo, and Q. Hu, "ECA-Net: Efficient channel attention for deep convolutional neural networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 11531–11539.
[50] Y. Cen et al., "Aerial hyperspectral remote sensing classification dataset of Xiongan New Area (Matiwan Village)," *J. Remote Sens.*, vol. 24, no. 11, pp. 1299–1306, 2020.
[51] Y. Zhong, X. Hu, C. Luo, X. Wang, J. Zhao, and L. Zhang, "WHU-hi: UAV-borne hyperspectral with high spatial resolution (H_2) benchmark datasets and classifier for precise crop identification based on deep convolutional neural network with CRF," *Remote Sens. Environ.*, vol. 250, Dec. 2020, Art. no. 112012.
[52] J. Jia et al., "Design, performance, and applications of AMMIS: A novel airborne multimodular imaging spectrometer for high-resolution Earth observations," *Engineering*, vol. 47, pp. 38–56, Apr. 2025.
[53] S. Kim, H. Wimmer, and J. Kim, "Analysis of deep learning libraries: Keras, pytorch, and MXnet," in *Proc. IEEE/ACIS 20th Int. Conf. Softw. Eng. Res., Manage. Appl. (SERA)*, 2022, pp. 54–62.
[54] Z. Xue, Y. Zhou, and P. Du, "S3Net: Spectral–spatial Siamese network for few-shot hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022.
[55] G. Mountrakis, J. Im, and C. Ogole, "Support vector machines in remote sensing: A review," *ISPRS J. Photogramm. Remote Sens.*, vol. 66, no. 3, pp. 247–259, May 2011.
[56] M. Belgiu and L. Drăguţ, "Random forest in remote sensing: A review of applications and future directions," *ISPRS J. Photogramm. Remote Sens.*, vol. 114, pp. 24–31, Apr. 2016.
[57] Y. Xu, L. Zhang, B. Du, and F. Zhang, "Spectral–spatial unified networks for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 10, pp. 5893–5909, Oct. 2018.
[58] W. Hu, Y. Huang, L. Wei, F. Zhang, and H. Li, "Deep convolutional neural networks for hyperspectral image classification," *J. Sensors*, vol. 2015, no. 1, pp. 1–12, 2015.
[59] A. Dosovitskiy et al., "An image is worth 16×16 words: Transformers for image recognition at scale," 2020, *arXiv:2010.11929*.
[60] Y. Chen, H. Jiang, C. Li, X. Jia, and P. Ghamisi, "Deep feature extraction and classification of hyperspectral images based on convolutional neural networks," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 10, pp. 6232–6251, Oct. 2016.
[61] N. Ma, X. Zhang, H. T. Zheng, and J. Sun, "ShuffleNet V2: Practical guidelines for efficient CNN architecture design," in *Proc. Eur. Conf. Comput. Vis.*, 2018, pp. 116–131.
[62] Y. Ma, B. Niu, and Y. Qi, "Survey of image classification algorithms based on deep learning," in *Proc. 2nd Int. Conf. Comput. Vis., Image, Deep Learn.*, Oct. 2021, p. 9.



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